

REVIEW

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Artificial photosynthesis for the production of value-added chemicals from CO₂ feedstock represents a promising approach toward achieving carbon neutrality. Nevertheless, the initial challenge lies in the capture and activation of CO₂, which is crucial for facilitating highly efficient conversion. Ionic liquids (ILs) are emerging as potential platforms for CO₂ capture and activation due to their diverse ion pairs, which assist in the synthesis and morphological control of photocatalysts, thereby exposing more active sites. Their tunable hydrophilicity and hydrophobicity create a confined environment around the catalyst, enabling selective control over the diffusion of reactants, intermediates, and products to the active sites. Additionally, the electronic structure of the photocatalyst can be adjusted through the careful selection of IL cations and anions, effectively reducing the activation energy barrier for CO₂ or optimizing the reaction pathway. Thus, three key roles of ionic liquids are thoroughly examined in this review. Furthermore, the underlying mechanism is also elucidated to facilitate a profound understanding of the photocatalytic CO₂ reduction process.

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Introduction

Excess emissions of CO₂ originating from the combustion of fossil fuels are the main cause of climate change. The concentration of CO₂ in the atmosphere has increased from approximately 280 ppm before the industrial era, to approximately 417 ppm by 2023. This growth trend correlates with the observed increase in global average temperature during the same period. Therefore, CO₂ is the primary greenhouse gas. In recent years, CO₂ capture and utilization (CCU) have attracted extensive attention.^{1–4} CO₂ can be efficiently converted into a wide variety of valuable chemicals including cyclic(poly) carbonates,⁵ carbon monoxide (CO), methanol (CH₃OH),

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methane (CH_4), *etc.* through thermal-catalytic, electrocatalytic, and photocatalytic processes.^{6–11}

Photocatalytic CO_2 conversion is a more promising technology because it does not require additional energy. Low reaction efficiency and poor product selectivity are the main obstacles to the practical application of photocatalytic CO_2 conversion technology. Moreover, the activation of CO_2 is challenging. Thus, the required CO_2 purity should be higher than 95% for laboratory-scale research, where the conversion rate can be measured. The integration of the CO_2 enrichment and transformation processes allows the direct utilization of diluted CO_2 , thereby eliminating the energy-intensive CO_2 compression step. Ionic liquids (ILs) hold promise as versatile media for the implementation of integrated capture and transformation technologies, given their capacity to act as sorbents for CO_2 , as well as solvents and supports in both homogeneous and heterogeneous catalysis.¹² Typically, IL aggregates are present instead of isolated ion pairs, even in the presence of co-solvents with high dielectric constants. This aggregation phenomenon significantly affects the physicochemical properties of the ionic liquid. ILs can act as nano-cages, offering a confined space. Transition metal complexes, biological, nanostructured, or traditional heterogeneous catalysts, can be readily immobilized within ILs and their hybrids, such as polymerized ILs or supported IL phases.¹³ The encapsulation of catalysts within the supramolecular structure of the IL creates a protective barrier around it, altering the geometric and electronic properties of the catalyst. This simple layer of ILs can also serve as

a physicochemical barrier, controlling substrate access, product release and the residence time of species that interact with catalytically active sites. Consequently, ILs are the most adaptable compounds for CO_2 capture, activation, and functionalization. The recent research progress in the application of ionic liquids (ILs) in the fields of heterogeneous catalysis,¹⁴ energy conversion and storage,^{15–17} biomedicine,¹⁸ gas separation and capture,¹⁹ and environmental remediation²⁰ has been summarized. These nice and timely reviews present the diverse functions of ILs and discuss the related mechanism. For catalysis, Qadir and Dupont²¹ gave a deep insight into the thermo- and photocatalytic activation of CO_2 in ILs. The tuning of the electronic properties of catalysts in ionic liquids (ILs) and the effects of ILs on the activation energy barrier of CO_2 were emphasized. However, the role of ILs in the photocatalytic CO_2 reduction reaction (CO_2RR) and the promoting effect of ILs on photocatalysts are still seldom summarized. Hence, a timely review is crucial at this stage to help researchers keep abreast of advancements and promptly recognize pertinent issues.

In this review, we discuss the role of ILs in the photocatalytic CO_2 conversion process and their impact on altering the electronic and geometric properties of photocatalysts. This overview aims to assist researchers in developing highly efficient photocatalysts for CO_2 reduction, and we anticipate that this work will open new avenues toward achieving net-zero CO_2 emissions.



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The research progress of ionic liquid assisted CO₂ photoreduction

Ionic liquids (ILs), as a class of room-temperature molten salts, are composed of asymmetric organic cations and inorganic anions. The excellent inherent features including dissolving/adsorbing ability, structural diversity, non-flammability and a wide electrochemical window endow ILs with various applications.²² In fact, in the field of photocatalytic CO₂ conversion, the imidazole-based ILs exhibit excellent CO₂ activation ability due to the formation of CO₂-imidazolium adducts.²³ Moreover, ILs also exhibit potential applications in photocatalysis, because some motifs in ILs are photoactive. Hence, it is necessary to summarize recent research progress on ILs applied in photocatalytic CO₂ conversion.

Photocatalytic CO₂ reduction represents a promising strategy for mitigating carbon dioxide emissions; however, it encounters several challenges, such as low reaction rates, poor selectivity, and catalyst deactivation. The incorporation of ionic liquids (ILs) into photocatalytic CO₂ reduction systems provides a multifaceted approach to enhancing the performance of CO₂ conversion processes. Specifically, ILs possess high ionic conductivity, which promotes charge transfer during photocatalytic reactions, thereby improving overall efficiency. The wide electrochemical stability window of ILs enables them to support various redox processes without undergoing decomposition, which is essential for maintaining stable photocatalytic activity. Furthermore, ILs show high CO₂ adsorption ability, significantly increasing the availability of CO₂ for reduction reactions and potentially stabilizing CO₂ in an activated form to improve reactivity. They can also improve the dispersion of catalysts, allowing better exposure of active sites and facilitating more efficient CO₂ reduction. Additionally, the tunable properties of ILs allow for the modulation of reaction pathways, directing the reduction process towards desired products and improving product selectivity. This review will be divided into three sections based on the different roles of ILs in CO₂ photoreduction.

Ionic liquid assisted synthesis of photocatalysts

As is well known, the poor light-harvesting ability, severe charge recombination, and sluggish kinetics of surface reactions are three problems that affect the efficiency of photocatalysts. Therefore, a variety of methods have been proposed to address these issues for improving photocatalytic activity. For example, morphology regulation, defect engineering, and element doping have been proven to be reliable strategies. ILs are considered multi-functional liquid materials, which have been widely used as green solvents for the assisted synthesis of nanomaterials, achieving the above goals. Recently, Adriana Zaleska-Medynska *et al.* summarized the effect of ILs on the surface and photocatalytic properties of semiconductors. The behavior of ILs in the assisted synthesis of semiconductor material particles and how their behavior affected the photocatalytic properties were discussed deeply.²⁴ Thus, the role of ILs in the synthesis of photocatalysts and the mechanism for

improving photocatalytic CO₂RR activity will be discussed in the following section, as presented in Fig. 1.

(1) Regulating the particle size: as is widely known, the exposure of more active sites as the particle size decreases can effectively enhance photocatalytic performance. The modification of metal clusters or single atoms can effectively improve the photocatalytic activity and selectivity for the CO₂ reaction by maximizing atomic utilization and altering the reaction path.^{25–28} However, the preparation of highly dispersed single atom photocatalysts is still challenging. ILs serving as green solvents can easily regulate the particle size and the morphology. The growth mechanism of nanomaterial assisted by using ILs is displayed in Fig. 2. In general, the morphology variation affected by the ionic liquid concentration is ascribed to the diffusion-limited aggregation process. The kinetics of the growth process are sensitive to organic or inorganic additives. ILs can produce bulky organic cations and small organic or inorganic anions, when they are homogeneously dissolved in the reaction mixture. The organic cation will react with the primary nuclei *via* hydrogen bonds or electrostatic force. The limitation in the diffusion of primary particles, along with the anisotropy in adsorption stability, leads to anisotropic particle growth and assembly into different morphologies. When the primary nuclei are surrounded by the IL domain, the organic cation is adsorbed onto a certain crystallographic plane more strongly than onto others, leading to the orientated growth/aggregation of primary particles. Hence, it is feasible to regulate the morphology of nanomaterials by changing the IL concentration, affecting the diffusion of primary particles. Albo *et al.* demonstrated that the 3-methyl-*n*-butylimidazolium tetrafluoroborate ([Bmim][BF₄]) ionic liquid acted as a template to control the size of Cu nanoparticles and a stabilizing agent to embed them on TiO₂ for efficient photoreduction of CO₂ to methanol.²⁹ The particle size of Cu ranges between 3 and 5 nm, and the Cu content was as high as 6.8%, indicating excellent dispersion of [Bmim][BF₄] in the synthesis of Cu nanoparticles (Fig. 2A). In addition, CuNi bimetallic nanoparticles with a size of 4–5 nm could also be synthesized with the assistance of [Bmim][BF₄] (Fig. 2B). The decoration of CuNi bimetallic nanoparticles enabled P25-TiO₂ to continuously produce methanol with a rate of 102.1 μmol g^{−1} h^{−1} from CO₂ photoreduction.³⁰ It is widely recognized that the catalytic activity of



Fig. 1 Role of ILs in the synthesis of photocatalysts.

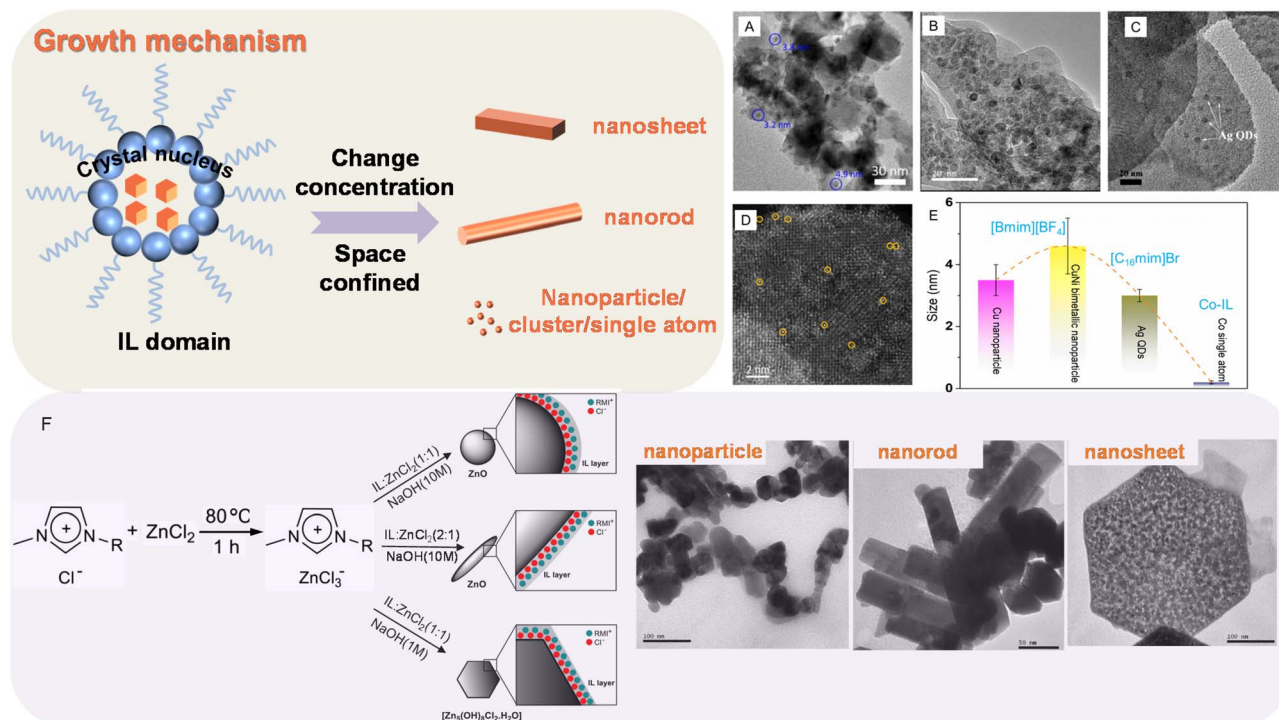


Fig. 2 The growth mechanism of IL-assisted synthesis of photocatalysts and the illustration of particle size regulation and morphology control by the addition of ILs. TEM images of (A) Cu nanoparticles, reproduced with permission from ref. 29, © Elsevier B.V. 2020. and (B) CuNi bimetallic nanoparticles synthesized by using [Bmim][BF₄], reproduced with permission from ref. 30, © Elsevier B.V. 2022. (C) Ag quantum dots (QDs) synthesized by using [C₁₆mim]Br, reproduced with permission from ref. 31, © Elsevier B.V. 2016. and (D) Co single atoms synthesized by using A336[CoCl₃], reproduced with permission from ref. 32, © American Chemical Society 2024. (E) The nanoparticle size regulated using different ILs. (F) Synthesis of trichlorozincate ILs and their use as precursors for the preparation of hybrid ZnO and [Zn₅(OH)₈Cl₂·H₂O] materials and their TEM images. Reproduced with permission from ref. 33, © American Chemical Society 2019.

metal nanoparticles is enhanced as their sizes diminish. Ag quantum dot (Ag QD) modified BiOBr ultrathin nanosheets were prepared using the 1-hexadecyl-3-methylimidazolium bromide ([C₁₆mim]Br) IL assisted solvothermal method (Fig. 2C). The evenly dispersed Ag QDs with the diameter less than 5 nm functioned as adsorption centers, charge separation sites and photocatalytic reaction hubs, thereby leading to improved photocatalytic activity.³¹ As metal nanoparticles shrink to the scale of individual atoms, some catalytic reaction pathways undergo a significant transformation, which has recently sparked a surge of interest in single-atom catalysis.^{34–36} The loading capacity and coordination mode of the single atoms affect the photocatalytic activity and product selectivity. However, increasing the content and tuning the coordination mode of single atoms *via* a simple method remain a challenge. Di *et al.* successfully synthesized Co single atom modified Bi₂₄O₃₁Br₁₀ atomic layers through a metal-based IL (Co-IL, A336 [CoCl₃]) induced strategy (Fig. 2D).³² Interestingly, the content of Co single atoms could be easily controlled by the addition of Co-IL. More importantly, the low-coordinated Co single-atoms loaded on the catalyst surface played a crucial role in decreasing the activation energy and rate-limiting step energy barriers, thereby enhancing the CO₂ reduction activity. Besides, the morphology of nanostructured materials could also be regulated by the addition of ILs. Zinc oxide can be synthesized

by the hydrolysis of imidazolium trichlorozincate ([Rmim][ZnCl₃]), where the shape can be tuned from irregular particles to nanorods and hexagonal-like nanosheets *via* increasing the IL amount, shown in Fig. 2F. As a result, the hybrid zinc oxide materials could promote the photoactivation of water and methane, achieving the H₂ yield of 1417 μmol g^{−1} h^{−1} and CO₂ yield of 67 μmol g^{−1} h^{−1}.³³

(2) Acting as a self-sacrificing template: ILs could also work as a self-sacrificing template for assisted synthesis. For instance, [C₁₆mim]Br IL not only acts as a self-sacrificing template and reactant to induce the formation of carbonized polymer dots (CPDs) and PbBiO₂Br, but also as a glue to *in situ* anchor CPDs on the PbBiO₂Br surface to form a heterojunction photocatalyst.³⁷ The introduction of CPDs effectively promotes light absorption, charge separation, and the adsorption/desorption equilibria of reactants/products on the photocatalyst surface. Thus, the obtained CPDs/PbBiO₂Br heterojunction photocatalyst exhibited a high selectivity, stability and enhanced CO evolution rate in water. Defect engineering is also a feasible strategy to improve the light-harvesting ability and charge separation of photocatalysts. Various photocatalysts with abundant defects such as ultrathin Bi₂₄O₃₁Cl₁₀ with surface oxygen vacancies,³⁸ bismuth-rich Bi₄O₅Br₂,³⁹ CdWO₄ with oxygen vacancies,⁴⁰ S-doped porous g-C₃N₄,⁴¹ P-doped g-C₃N₄ (ref. 42), *etc.* have been prepared using the IL-assisted

synthetic method for CO₂ photoreduction⁴³ and other photocatalytic applications.^{44–49} Besides, some bi-functional ionic liquids could serve as both the intercalator and stabilizer to strip the two-dimensional imine-linked covalent organic framework (COF) in water for efficient photocatalytic CO₂ conversion to CO.⁵⁰ In addition to regulating the nanoparticle size and morphology of photocatalysts, ILs also play a critical role in forming highly active photocatalytic structures such as metalated porphyrins, which we explore next.

(3) Assisting the formation of metalated porphyrin: studies have shown that porphyrin-based metal-organic frameworks (PMOFs) are a promising platform for the design of artificial photosynthetic systems.⁵¹ Moreover, the MN₄ coordination is beneficial for high affinity toward CO₂ molecules and can be formed by the incorporation of metal ions into the porphyrin ligand. The metalation of the porphyrin macrocycle is quite important in modifying the electronic structures of PMOFs, not only providing atomically dispersed catalytic centers, but also boosting the charge separation efficiency.^{52–54} Of note, a novel two-dimensional porphyrin-based Mn-MOF with high loading (67 wt%) of Mn(II)-metalated porphyrin (Mn-TCPP) induced by the [Bmim]Br ionic liquid was synthesized. The atomically dispersed porphyrin Mn(II) sites integrated CO₂ adsorption sites and active centers in Mn-MOF photocatalysts, and the Mn-TCPP ligands also worked as visible-light-harvesting units to generate electron-hole pairs. These advantages endowed the Mn-MOF with the overall photocatalytic conversion of CO₂ to CH₄ under gas-solid conditions.⁵⁵ This work provided a feasible strategy to fabricate MOF photocatalysts with a high loading of porphyrin metal sites induced by ionic liquids for efficient CO₂ photoconversion. Similarly, an anionic porous Zn-metalated porphyrin MOF induced by [Bmim]Br was also successfully synthesized and exhibited a highly selective photocatalytic CO₂ conversion to CH₄ (~70%) in H₂O vapor.⁵⁶

In summary, ILs exhibited diverse functions to assist the synthesis of photocatalysts for improving CO₂RR activity throughout: (1) affecting the particle size to expose more active sites and shorten the diffusion paths of charge carriers; (2) creation of new energy states or surface defects to reduce bandgap and/or inhibit charge recombination and/or provide additional active sites; (3) increasing the loading of porphyrin metal sites to provide more active sites.

Ionic liquids as reaction media

For the CO₂ conversion, the biggest challenge is activation of CO₂ owing to its high thermodynamic stability. The versatility of ILs as both synthesis agents and reaction media underscores their central role in CO₂ photoreduction systems. Natural photosynthesis provides a source of inspiration for mild catalytic conversion of CO₂. The synergetic effect between photocatalysts and reaction media is highly desirable to boost the photocatalytic conversion of CO₂. ILs, as typical examples of such collaborative media, have the remarkable ability to absorb high concentrations of CO₂ through both physisorption and chemisorption and thus co-catalyze CO₂ conversion. In particular, the chemical reaction of many ionic liquids with CO₂

resembles the first step of CO₂ activation in the plant photosynthesis cycle.²¹ For example, 1-aminopropyl-3-methylimidazolium bromide ([APMIm]Br) was selected as the electrolyte to mimic the plant environment for CO₂ reduction to methanol by using an artificial photosynthesis cell, where BiVO₄ worked as the photoanode and Mn-doped CdS and CdSeTe QD cosensitized TiO₂ worked as a photocathode.⁵⁷ First, an [APMIm-CO₂]⁺ complex was generated quickly through a hydrogen bonding interaction between CO₂ and the [APMIm]⁺ cation in the [APMIm]Br IL electrolytes (Fig. 3A). Under light illumination, the photoexcited electrons reacted with activated CO₂ and active hydrogen atoms. In this process, [APMIm]Br IL served as a cocatalyst, which could decrease the activation barrier and reduce the overpotential requirement for the CO₂-to-hydrocarbon conversion. Finally, this system showed a high efficiency in yielding methanol at a rate of 90 μM h⁻¹ cm⁻² at -0.9 V versus the saturated calomel electrode under 200 mW cm⁻² irradiation. Furthermore, [APMIm]Br was also used as an absorbent and electrolyte in the WO₃/BiVO₄

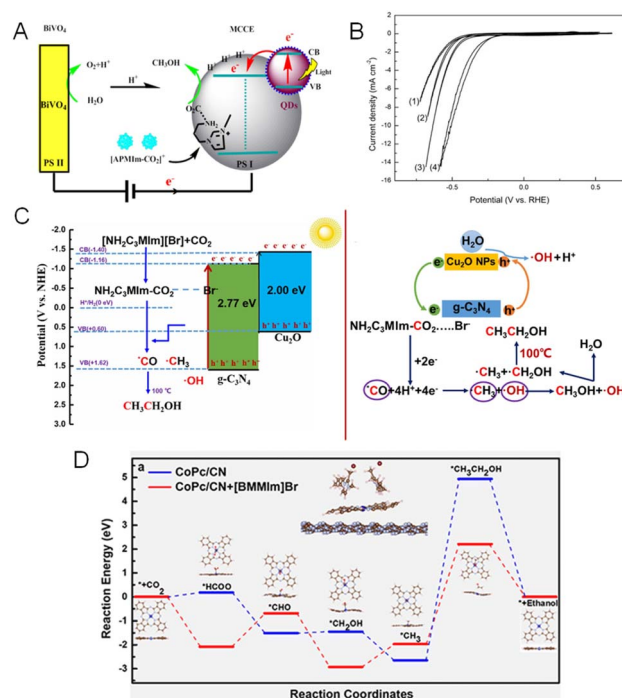


Fig. 3 The promoting effect of ILs on photoelectrocatalytic CO₂ reduction when ILs served as reaction media. (A) Proposed mechanism of photoelectrocatalytic CO₂ reduction in the BiVO₄/[APMIm]Br IL system, Reproduced with permission from ref. 57, © Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim 2018. (B) CVs recorded in aqueous solutions of [APMIm]Br, [C₄MIm]Br, and KHCO₃. Conditions: (1) CO₂ + KHCO₃ electrolyte, (2) CO₂ + [C₄MIm]Br electrolyte, (3) CO₂ + [APMIm]Br electrolyte, and (4) N₂ + [APMIm]Br electrolyte, Reproduced with permission from ref. 58, © Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim 2017. (C) Proposed mechanism of photothermal catalytic conversion of CO₂ to ethanol and the possible pathway over Cu₂O/g-C₃N₄ in [APMIm]Br electrolyte, Reproduced with permission from ref. 59, © Elsevier B. V. 2021. (D) Free energy diagrams for CO₂ reduction on CoPc/CN in aqueous [BMMIm]Br solution, Reproduced with permission from ref. 60, © The Royal Society of Chemistry 2023.

photoelectrochemical cell (PEC) system.⁵⁸ As shown in Fig. 3B, cyclic voltammetry (CV) curves were recorded to study the effect of [APMIm]Br on the reduction of CO₂. It was obvious that the onset potential of the CO₂-purged aqueous [APMIm]Br solution shifted anodically compared with KHCO₃ solution, indicating that the overpotential of CO₂ reduction was lowered in [APMIm]Br solution. This was ascribed to the formation of the [APMIm-CO₂] complex. Herein, [APMIm]Br played a critical role in promoting the conversion of CO₂ to formic acid and suppressing the production of H₂. As a result, the faradaic efficiency for formic acid production was as high as 94.1% at an applied voltage of 1.7 V. In the Cu₂O/g-C₃N₄ system, [APMIm]Br showed superior CO₂ reduction performance as an electrolyte compared with KHCO₃. By coupling thermal field and ionic liquid, the yield of ethanol reached 0.71 mmol g⁻¹ h⁻¹ over the Cu₂O/g-C₃N₄ heterojunction photocatalyst.⁵⁹ Undoubtedly, [APMIm]Br successfully suppressed the H₂O reduction to H₂. Furthermore, the formation of the [APMIm-CO₂] complex provided a low energy pathway to activate CO₂, facilitating the continuous C-C coupling under light irradiation. It was also an important reason for increasing the yield of ethanol, and the mechanism is displayed in Fig. 3C. Interestingly, [BMMIm]Br exhibited a similar promotional effect for CO₂ reduction to methanol in the g-C₃N₄/CoPc PEC system.⁶⁰ As seen in Fig. 3D, the presence of ILs obviously promoted C-C coupling and stabilized the *HCOO intermediate, thereby increasing the yield of methanol and promoting the formation of ethanol. In conclusion, ILs played dual roles as both electrolytes and stabilizers. The [BMMIm]Br ILs greatly improved charge separation, selectively stabilized *HCOO intermediates, and significantly reduced the Gibbs free energy of the rate-determining step of ethanol formation. In the Cu₂O/TiO₂ nanoarray PEC system, a strong interaction between 1-ethyl-3-methylimidazolium tetrafluoroborate ([Emim]BF₄) and catalyst surface greatly enhanced charge separation and promoted C-C coupling, effectively improving the selective production of ethanol.⁶¹

Metal salts are the widely used reaction media or the major component of electrolytes, while ILs only serve as additives. Thus, it is necessary to investigate the effect of IL content and composition in the electrolyte solution on the photocatalytic activity of CO₂ conversion. In the (Mo-Bi)S_x modified mesoporous CdS photocatalytic system, [Bmim]BF₄ was added into acetonitrile as a co-solvent for the CO₂ photochemical reduction reaction. In this reaction, HCOOH was the only carbonaceous product and H₂ was the only byproduct. Moreover, the H₂ evolution rate decreased, whereas the HCOOH formation rate sharply increased with increasing the content of [Bmim]BF₄. In this report, [Bmim]BF₄ not only reduced the activation energy of CO₂ *via* the formation of a weak complex of (Bmim-CO₂-BF₄), but also enhanced charge separation.⁶² In the S-scheme BiVO₄@TiO₂ nanoglass/needle array photocatalytic system, only 5% [Bmim]BF₄ was added, and the CH₄ selectivity increased to 57.3% (132.7 μmol m⁻² h⁻¹). The author refrained from delving excessively into the enhancement of photocatalytic CO₂ reduction activity by optimizing the ionic liquid concentration, and considered the improvement was attributed to the CO₂ activation promoted by the assistance of [Bmim]BF₄.⁶³ In

the STO/Cu@Ni/SiO₂/TiN heterojunction photocatalytic system, [Bmim]Br served as the primary reaction media, and CO₂ was bubbled into the reactor. The ionic liquid in this gas-solid-liquid system could activate CO₂, resulting in a high evolution rate of 25.8 μmol g⁻¹ h⁻¹ for methanol production, which is 50 times that of a traditional gas-solid system.⁶⁴ In a photochemical tandem system, the promotional effect of [Emim]BF₄ on CO₂ photofixation in various solvents including H₂O, DMF (*N,N*-dimethylformamide), MeCN (acetonitrile), THF (tetrahydrofuran), and BTF (benzenyltrifluoride) was investigated in detail. In Fig. 4A, it was obvious that all solvent systems displayed enhanced photocatalytic activity towards CO₂ reduction upon adding [Emim]BF₄. Among these, the yield of CO and H₂ reached the highest in the IL-MeCN mixture with a ratio of 75% (v/v). As displayed in the inset of Fig. 4A, the production of both CO and H₂ increased with increasing ratio of [Emim]BF₄/H₂O, but it decreased sharply when H₂O was removed, indicating that the mutual action of IL and H₂O was the key to improve the photocatalytic activity toward CO₂ reduction. In addition, various ILs were also employed to investigate the effect of the counterions and the substituents of the organic components on CO₂ photoreduction. As elucidated in Table 1, the Tf₂N⁻ anion showed the best promotional effect in the production of CO and H₂, which was ascribed to the low viscosity. In summary, the promotional effect of ILs on CO₂ photoreduction was closely related to the chemical properties of counterions and the organic functional groups on the imidazolium cation.⁶⁵ Lin *et al.* also studied the relationship between CO₂ reduction reaction (CO₂RR) activity and IL concentration in the [Emim]BF₄ + TEOA + H₂O system, where CdS was employed as a photocatalyst. They found that the viscosity and conductivity of reaction media affected the CO₂RR activity and selectivity. However, the underlying reason was not clear.⁷¹ Prashant K. Jain *et al.* systematically investigated the relationship between photocatalytic CO₂ reduction activity of Au nanoparticles and IL concentration. The CO₂ reduction activity exhibited a volcano relationship as a function of the [Emim]BF₄ concentration (Fig. 4B), and the selectivity for C₂₊ production was ~50% in 1–10 mol% [Emim]BF₄ solution. The synergistic effect of plasmonic and IL promoted the production of C₁–C₃ hydrocarbons.⁶⁶ The selectivity of hydrocarbon products was closely related to the HER branch and CO₂RR branch, which was regulated by the [H⁺] concentration of the reaction medium (Fig. 4C). In an IL-containing medium, the CO₂RR was inherently the dominant reaction branch for accepting the plasmonically generated electrons, while the HER was the secondary branch. As a result, increasing H⁺ availability to an optimal value enhanced hydrocarbon selectivity while suppressing the HER.⁶⁷ The synergic effect of Au nanoparticles and ILs to enhance CO₂ conversion into value-added products was also observed in the TiO₂ nanotube photocatalytic system. The addition of [Bmim]BF₄ enhanced the solubility of CO₂ in water, and the surface plasmon resonance of Au nanoparticles improved the photoexcitation. Thus, the methanol generation increased from 186.8 to 320.8 μM and methane from 67.6 to 159.2 μM, respectively, when the content of [Bmim]BF₄ was increased from 1% to 4%.⁷²

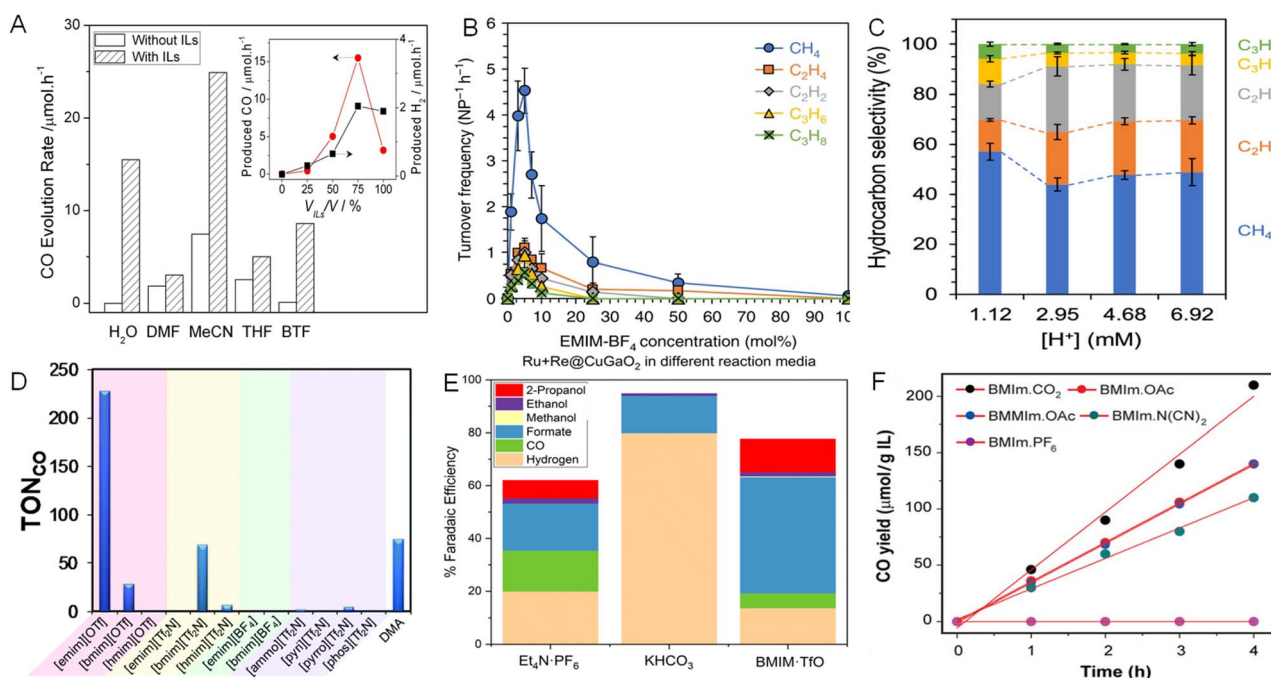


Fig. 4 The promoting effect of ILs on the photocatalytic CO₂ reduction reaction when adding ILs into reaction media. (A) The comparison of photocatalytic CO₂ reduction activity in different solvents with/without ILs, reproduced from an open access ref. 65. (B) Turnover frequencies of hydrocarbon products formed in the CO₂RR plotted as a function of the [Emim]BF₄ concentration (mol%), reproduced from an open access ref. 66. The CO₂ conversion activity peaks at 5 mol% of [Emim]BF₄. (C) Selectivity of hydrocarbon products generated in the CO₂RR as a function of the [H⁺] in 5 mol% [Emim]BF₄. Reproduced with permission from ref. 67, © American Chemical Society 2019. (D) TON_{CO}s of the photocatalytic reactions using 0.5 mM Ir, 0.1 M BIH and 0.05 mM Re(Br) in various ionic liquids, Reproduced with permission from ref. 68, © The Royal Society of Chemistry 2020. (E) faradaic efficiencies of gas and liquid products produced in 120 min chronopotentiometry at -1.5 mA·cm⁻², under continuous simulated sunlight irradiation, using different electrolytes, 0.3 M Et₄N·PF₆ in acetonitrile, 0.1 M KHCO₃ in H₂O and 0.3 M BMIM·TfO in acetonitrile, all saturated with CO₂. Reproduced with permission from ref. 69, © The Royal Society of Chemistry 2023. (F) Photocatalytic reduction of CO₂ by different ILs. Reaction conditions: IL (5 mg), CO₂ (atmospheric pressure), H₂O (3 mL), temperature (25 °C), and Xe arc lamp (300 W). Reproduced with permission from Ref. 70, © Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.

To avoid the mass transfer issue induced by high viscosity of ILs, a set of imidazolium-based ILs with variable counterions were used as a low-level additive (0.5–20% (w/v)) in three aprotic solvents (DMF, MeCN and DMSO) for understanding the cooperative effect and counterion effect on CO₂ photoreduction.⁷³ Therein, [Re(bpy)(CO)₃]Cl and [Ru(bpy)₂](PF₆)₂ served as a catalyst and a sensitizer, respectively. In this system, it was found that the increase of [C₂mim][OAc] content from 0 to 10% (w/v) in MeCN resulted in a stepwise increase of CO formation and successively improved H₂ suppression, whereas a higher content was not beneficial. The selectivity toward CO formation was also reduced due to the presence of the HCO₃⁻ species when water was added. More importantly, the impact of different counterions could be correlated with the Kamlet-Taft parameter β. Ionic liquids with high nucleophilicity and high β value increased CO formation. Except for imidazolium-based ILs, other kinds of ILs also exhibited a positive effect in promoting photocatalytic CO₂ reduction. For instance, Asai *et al.* systematically compared the CO₂ photoreduction ability in twelve kinds of ILs, using an Ir-based sensitizer and [Re(bpy)(CO)₃]Cl as a catalyst. The effect of molecular structures of ILs on photosensitizing and catalytic cycles was studied. Some guidelines for the selection of ionic liquids were obtained.

The ionic liquids with [BF₄]⁻ anions or [phos]⁺ cations were not suitable for the photocatalytic reactions, whereas [Imidazolium][OTf]-type ionic liquids were identified as most efficient solvents for photocatalytic CO₂ reduction, shown in Fig. 4D. The study also emphasized the negative impact of conventional additives on the durability of photocatalytic reactions in ionic liquids, suggesting that the use of additives may not provide substantial help when the photocatalytic ability is insufficient.⁶⁸ The proposed guideline for the selection of ionic liquids is valuable. Bernados *et al.* synthesized a CuGaO₂-based hybrid supramolecular photocathode and revealed the beneficial effect of 1-butyl-3-methylimidazolium triflate ([Bmim][TfO]) on the conversion of CO₂ to formic acid and the suppression of H₂ production. Compared with [Et₄N]PF₆ and KHCO₃, a higher selectivity to CO₂ reduction products was obtained with formate as the major product in [Bmim][TfO] (Fig. 4E).⁶⁹ The impact of ILs on the photocatalytic performance was tested in the CuPS/Fe systems by Jennifer Strunk. Different ILs including an imidazolium-, a pyrrolidinium-, and a phosphonium-based IL were used as solvents for photocatalytic CO₂ reduction. Among them, 1-butyl-1-methyl-pyrrolidinium dicyanamide ([Bmppyrr][N(CN)₂]) was found to be the most suitable reaction medium for the selective production of CO. Unfortunately, the

Table 1 The effect of various ILs on the photocatalytic conversion of CO₂ to CO under mild conditions

IL	CO/ μmol	H ₂ / μmol	CO + H ₂ / μmol	Sel./%
[EMIM][Tf ₂ N]	34.7	10.9	45.6	76.1
[EMIM][OTf]	3.7	1.1	4.8	77.1
[EMIM][OTf]	0.4	0.1	0.5	80.0
[EMIM][Ac]	n.d.	n.d.	—	—
[EMIM][DCA]	0.1	0.2	0.3	33.3
[EMIM][BF ₄]	31.0	4.1	35.1	88.3
[BMIM][BF ₄]	28.0	1.6	29.6	94.6
[HMIM][BF ₄]	13.1	0.5	13.6	96.3
[OMIM][BF ₄]	6.8	0.3	7.1	95.8
[BDIMIM][BF ₄]	13.7	1.1	14.8	92.3

underlying mechanism was not revealed.⁷⁴ Stéphane Daniele *et al.* studied the effect of several types of co-solvents such as H₂O, isopropanol (iPrOH), and 1-(2-hydroxyethyl)-3-methylimidazolium bis(trifluoromethanesulfonyl)imide [C₁C₂(OH)ImNTf₂] on the CO₂RR of Pd/g-C₃N₄ in 1-butyl-3-methylimidazolium acetate (C₁C₄ImOAc). After screening, iPrOH displayed the best performance in the conversion of CO₂ into CO, followed by C₁C₂(OH)ImNTf₂, and it was found that H₂O was detrimental. This is because iPrOH allowed ILs to preserve the pre-activation of CO₂ while acting as a hole scavenger to avoid charge recombination.⁷⁵ This work provided new insight into the photocatalytic conversion of CO₂ in ILs and the importance of co-solvents. Then, Stéphane Daniele *et al.* further investigated the photocatalytic reduction of CO₂ in iPrOH or H₂O and in the presence of imidazolium-based ILs with different anions. Some similar conclusions could be obtained. The CO production exhibited a positive relationship with the light-harvesting ability and Kamlet-Taft parameter β , which was related to the cations and the anions of ILs, respectively. In iPrOH, the ability of the anion to interact with the solvent could be reflected by the Kamlet-Taft parameter β . The higher the value of β was, the more efficient the CO production was. In aqueous media, pH was the key to affecting the selectivity. When pH > 8, carbonate ions detrimental to CO₂ photoreduction formed.⁷⁶

Of note, the formation of the imidazolium-CO₂ complex is the key to improving the photoreduction of CO₂ in ILs. Thus, the activation path of CO₂ in imidazolium-based ILs should be explored. In the photolysis process of 1-*n*-butyl-3-methylimidazolium-2-carboxylate (BMIm.CO₂) IL, the BMIm radical cation ([BMIm]^{•+}) and the CO₂ radical anion ([CO₂]^{•-}) were detected by ESR and NMR. Thus, the activation path proceeded through homolytic cleavage of the imidazolium-CO₂ adduct was confirmed. Then, photocatalytic CO₂ reduction in different ILs was compared under the same reaction conditions, and the results are displayed in Fig. 4F. Apparently, no production of CO was detected in BMIm⁺PF₆. This may be correlated with the basicity of the anion, which affects the adsorption of CO₂. Therefore, the probable limiting step was the formation of the imidazolium-CO₂ adduct, which facilitated CO₂ adsorption and activation by reducing the potential for the formation of the CO₂ radical anion.⁷⁰ The positive role of ILs in improving photocatalytic CO₂ reduction activity was further proved by Cronin's work.⁷⁷ The photoelectrochemical CO₂ reduction to CO over the TiO₂ layer deposited on the surface of the InP photocatalyst was carried out in a nonaqueous solution consisting of [Emim]BF₄ and acetonitrile. The faradaic efficiency of CO₂ reduction reached 99% at an underpotential of +0.78 V. Moreover, NMR spectra showed that the [Emim]⁺ ions in the solution formed an intermediate complex with [CO₂]^{•-}, thereby lowering the reaction energy barrier.

In conclusion, the molecular structure, basicity of counterions, viscosity, and the Kamlet-Taft parameter β must be considered for the selection of ILs as reaction media for CO₂ photoreduction. Besides, the hydrogenation step in the production of hydrocarbons and competitive hydrogen evolution reaction are affected by the H⁺ concentration of reaction media, which can be regulated by the IL concentration.

Table 2 The frequently used ILs and their properties

Name	Structure	Formular	Molecular weight	Viscosity (Pa s, 298.15 K)	Conductivity (S m ⁻¹ , 298.15 K)	CAS registry number
1-Butyl-3-methylimidazole tetrafluoroborate ([Bmim]BF ₄)		C ₈ H ₁₅ BF ₄ N ₂	226.03	0.308 ± 0.022	0.332 ± 0.01	174501-65-6
1-Butyl-3-methylimidazole hexafluorophosphate ([Bmim]PF ₆)		C ₈ H ₁₅ F ₆ N ₂ P	284.19	0.271 ± 0.062	0.1379 ± 0.0083	174501-64-5
1-Butyl-3-methylimidazolium trifluoromethanesulfonate [Bmim][OTf]		C ₉ H ₁₅ F ₃ N ₂ O ₃ S	288.29	0.0991 ± 0.0051	0.29 ± 0.003	174899-66-2
1-Butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide [Bmim][Tf ₂ N]		C ₁₀ H ₁₅ F ₆ N ₃ O ₄ S ₂	419.36	0.0511 ± 0.0017	0.387 ± 0.049	174899-83-3
1-Ethyl-3-methylimidazole acetate ([Emim]OAc)		C ₈ H ₁₄ N ₂ O ₂	170.21	0.1032 ± 0.0054	0.278 ± 0.028	143314-17-4
1-Ethyl-3-methylimidazole tetrafluoroborate ([Emim]BF ₄)		C ₆ H ₁₁ BF ₄ N ₂	197.97	0.0434 ± 0.0047	1.394 ± 0.029	143314-16-3

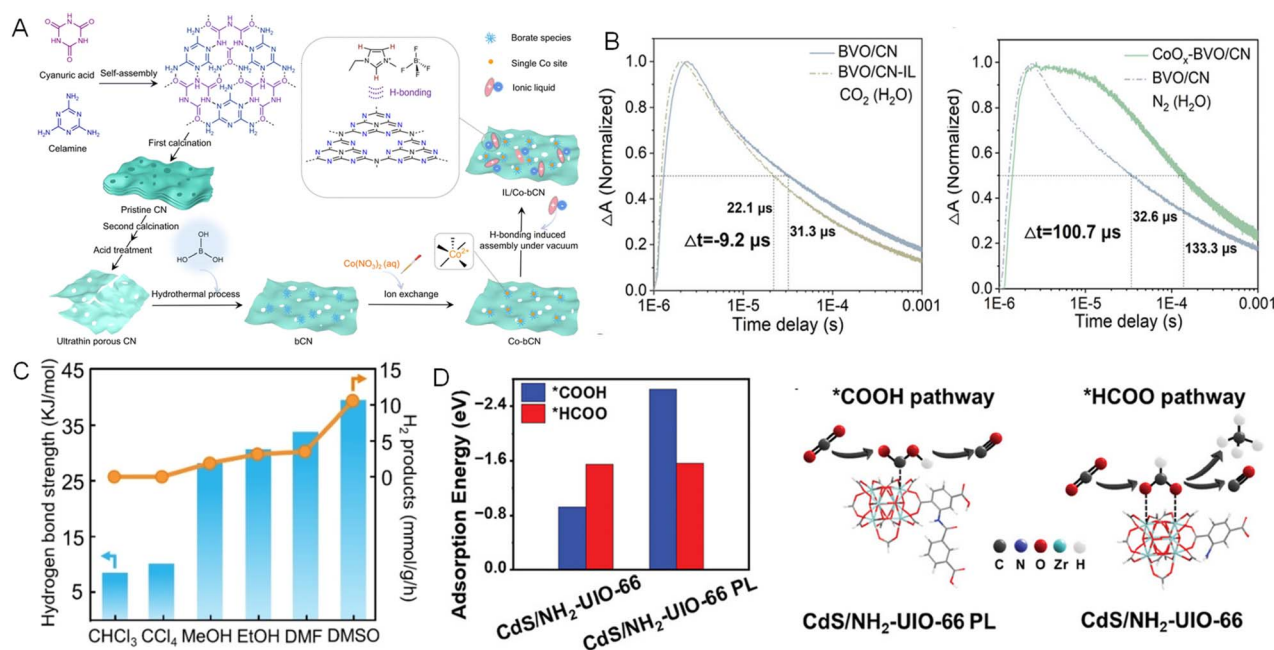


Fig. 5 ILs served as cocatalysts for boosting charge separation. (A) Illustration of the synthetic procedures of IL/Co-bCN photocatalysts, reproduced from an open access ref. 78. (B) μ s-TAS decay kinetics of BVO/CN and BVO/CN-IL under CO₂(H₂O) conditions, and μ s-TAS kinetics of BVO/CN and CoO_x-BVO/CN under N₂(H₂O) conditions, Reproduced with permission from ref. 79, © Wiley-VCH GmbH 2023. (C) Comparison of the hydrogen production rate and hydrogen bond strength with various additives, Reproduced with permission from ref. 80, © Wiley-VCH GmbH 2024. (D) The calculated adsorption energy between different intermediates and various photocatalysts, and schematic diagram of the CO₂ adsorption mode and main possible reaction pathways for CdS/NH₂-UIO-66 and CdS/NH₂-UIO-66 PL. Reproduced with permission from ref. 81, © Wiley-VCH GmbH 2024.

Therefore, the selection and concentration of ILs are both considered for constructing highly efficient photocatalytic systems. Finally, the frequently used ILs as reaction media for CO₂ reduction are listed in Table 2.

Ionic liquid as a mediator

Designing excellent photocatalysts for artificial photosynthesis to reduce CO₂ into fuels or high value-added chemicals is highly desirable. However, most reported semiconductor-based photocatalysts cannot capture CO₂ directly. Thus, different methods have been developed for the construction of surface structures and the design of active sites. ILs present excellent CO₂ adsorption ability due to their versatile interaction mode with CO₂, including physical adsorption, weak interaction and chemical interaction. Moreover, the tunable structure and diverse composition enable ILs to act as functional species to modify host materials to substantially enhance their CO₂ capture ability. Imidazolium-based ILs, a class of widely used ILs, have been employed as cocatalysts in electrochemistry. The utilization of imidazolium-based ILs could lower the potential for the formation of *CO₂[−] intermediates to promote the CO₂ conversion rate and then improve faradaic efficiencies. Importantly, the C(2)-H in the imidazole ring tends to complex with CO₂ instead of H₂O, thus inhibiting the competitive hydrogen evolution and being conducive to CO₂ reduction. The imidazolium ring can also provide linkage sites for the connection with nanomaterials by H-bonding interactions. For example, Jing *et al.* reported that [Emim][BF₄] and Co single atoms were co-loaded on CN nanosheets for CO₂ photoreduction. Herein, the [Emim][BF₄] was connected with an amino group of CN through H-bonding interactions, as displayed in Fig. 5A. The ILs could accept the electrons generated from CN and serve as active sites for CO₂ reduction. On the other hand, Co single atoms could trap photogenerated holes for water oxidation. Thus, the spatially separated reduction domains and oxidation domains resulted in fast electron kinetics and nearly 100% selectivity towards CO₂ reduction.⁷⁸ Besides, Jing's group also employed [Emim][NTf₂] and CoO_x clusters as spatially separated redox sites to modify the BiVO₄/g-C₃N₄ Z-scheme heterojunction (BVO/CN). Herein, [Emim][NTf₂] connected with g-C₃N₄ through H-bonding interactions and could accept photo-excited electrons to activate the adsorbed CO₂. To understand the effect of IL on the electron kinetics, the *in situ* μs-TAS decay kinetics of BVO/CN and IL modified one were recorded and shown in Fig. 5B. It was obvious that the electron $t_{50\%}$ in BVO/CN-IL was shortened by 9.2 μs compared with that of BVO/CN under the CO₂(H₂O) conditions, indicating that the IL modification was beneficial for accelerating the electron kinetics for CO₂ reduction in a CO₂-enriched environment. On the other hand, the electron $t_{50\%}$ in BVO/CN was dramatically prolonged after CoO_x cluster decoration in the N₂(H₂O) system. Thus, the spatially separated redox site modified BVO/CN Z-scheme heterojunction exhibited a high electron transfer efficiency, reaching up to 36.4% for CO₂ reduction.⁷⁹ Notably, the H-bonding interactions are not always beneficial to the improvement of photocatalytic activity. For instance, Deng *et al.* found

that the breakdown of the hydrogen bond network in the reaction system can suppress H₂ evolution, thereby improving CO₂ photoreduction performance.⁸⁰ In this work, photosensitive poly(ionic liquid)s were designed and utilized as catalysts. The strength of the hydrogen bond network with various additives could affect the proton transfer behaviours, as shown in Fig. 5C. The catalyst system containing CHCl₃ as an additive presented the highest CO production rate of 35.4 mmol g^{−1} h^{−1} and boasted an impressive CO selectivity of 98.9%. Thus, breaking the hydrogen bond network effectively inhibited H₂ evolution and accelerated CO₂ conversion.

In addition to H-bonding interactions, different chemical bonds were also reported for linkage. The amide bonds in a porous liquid (NH₂-UIO-66 PL), which was composed of NH₂-UIO-66 and [M2070]IPA, exhibited an excellent ability to facilitate the adsorption and activation of CO₂. As shown in Fig. 5D, the Zr⁴⁺ sites in CdS/NH₂-UIO-66 preferred to adsorb the *HCOO intermediate compared to the *COOH intermediate. In contrast, the presence of amide bonds in CdS/NH₂-UIO-66 PL significantly enhanced the adsorption energy of *COOH on the Zr⁴⁺ sites. In addition, the DFT calculations demonstrated that the energy barrier for the generation of *COOH species (+0.08 eV) over CdS/NH₂-UIO-66 PL was much lower than the formation of the *HCOO intermediate (+1.39 eV). Herein, the introduced amide moieties not only enriched electron density at active sites but also stabilize the *COOH intermediate, resulting in the high-performance of CO₂-to-CO photoreduction with 100% selectivity in the CdS/NH₂-UIO-66 PL hybrid catalyst.⁸¹ The tunable alkyl chain structure endowed imidazolium-based ILs with diverse chemical connection modes. Thiol-modified imidazolium compound MEMI served as a capping ligand connected with ZnSe QDs *via* a thiol group, and the strong interaction between MEMI and the QDs was corroborated by many advanced characterization studies. The imidazolium capping ligand stabilized the surface-bound *CO₂[−] intermediate, increasing the yield and selectivity toward CO production.⁸² Other than connecting with the cations of ILs, the anion part could also bind with the surface of photocatalysts. For example, [Bmim][BF₄] connected with the Chl(Cu)/TiO₂ surface through a Ti–F bond. Hence, a chloroplast-inspired inverse opal structured IO-TiO₂/Chl/IL photocatalyst was designed for CO₂ photoreduction. Therein, [Bmim][BF₄] resembled the CO₂ capture function of stroma in chloroplast. And the Ti–F bond significantly promoted electron transfer ability, leading to excellent photocatalytic performance (44.9 μmol g^{−1} h^{−1}) with high CH₄ selectivity (93%).⁸³ Recently, Zhao *et al.* synthesized the Ti–F bridged IL-CuCQDs-F/TiO₂ inverse opal composite (IO-CFTi) by using ethylenediaminetetraacetic acid (EDTA(Cu)) and imidazole ionic liquid 1-(2-Hydroxyethyl)-3-methylimidazolium tetrafluoroborate ([HOEtMIM][BF₄]). The Ti–F bonding interface of IO-CFTi assisted photogenerated electron transportation from TiO₂ to CO₂ for its reduction. Consequently, IO-CFTi achieved a remarkably increased CO production rate of 78.1 μmol g^{−1} h^{−1} with 98% selectivity. Apparently, it is an intriguing and promising strategy to design different chemical interactions to link ILs and photocatalysts. The strong chemical

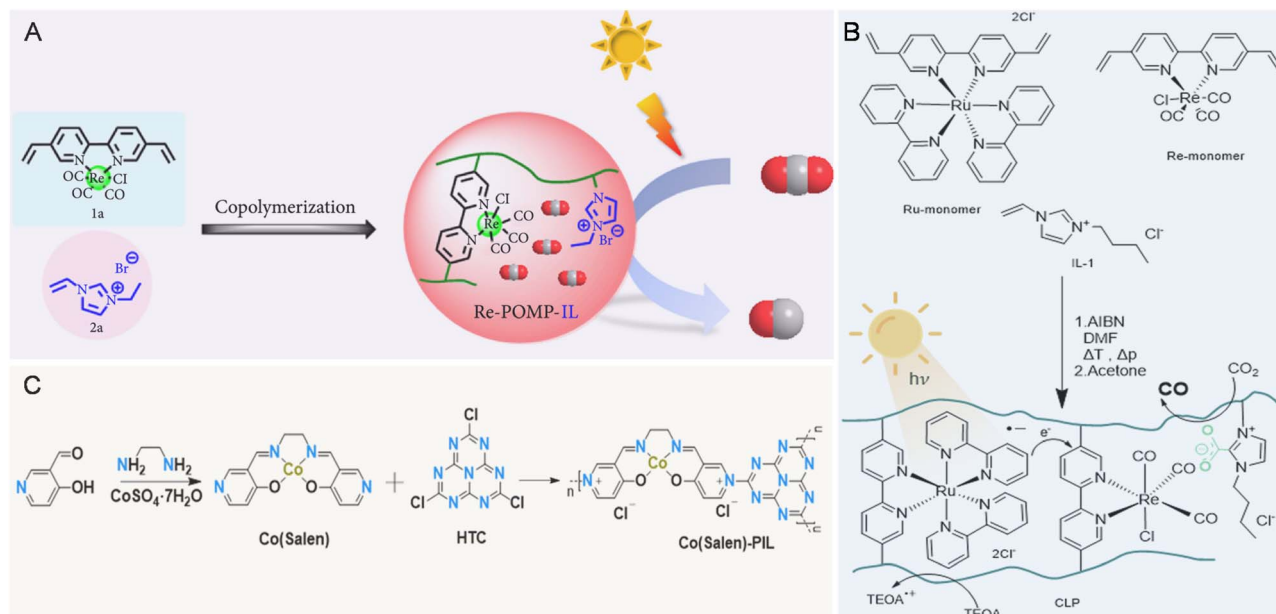


Fig. 6 The synthesis process and the possible molecular structure of IL-modified photocatalyst. (A) The IL-modified Re-based POMP for CO₂ photoreduction to CO, reproduced from an open access ref. 84. (B) Schematic representation of monomers used for the radical polymerization of crosslinked polymers (CLP) with AIBN (azobisisobutyronitrile) towards a polymer framework for photocatalytic CO₂ reduction, reproduced with permission from ref. 85, © The Royal Society of Chemistry 2024. (C) Scheme showing the synthetic strategy for Co(Salen)-PIL, reproduced with permission from ref. 86, © The Royal Society of Chemistry 2024.

interactions not only stabilize the linkage, but also provide an electron transfer path, effectively promoting charge separation.

Besides, the chemical bonding between ILs and semiconductor photocatalysts was not pointed out, and the photocatalytic CO₂RR activity was also enhanced after the modification of ILs. In particular, the IL-modified Re-based porous organometallic polymers (Re-POMP-IL) could be obtained through radical copolymerization of [(5,5'-divinyl-2,2'-bipyridine)Re(CO)₃Cl] and 3-ethyl-1-vinyl-1H-imidazol-3-ium bromide for CO₂ reduction to CO, shown in Fig. 6A. The Re-POMP-IL photocatalyst showed better performance than Re-POMP in promoting CO₂ to selectively produce CO, and the CO selectivity was up to 95.5%, which was ascribed to the high CO₂ affinity of IL.⁸⁴ Coincidentally, Bica-Schröder also reported that the IL-based polymer was prepared by radical copolymerization of 1-butyl-3-vinylimidazolium chloride and photocatalytically active Ru- and Re-monomer for highly selective CO production (Fig. 6B). The Ru-complex, Re-complex, and imidazolium core units in the polymeric material served as the CO₂ reduction catalyst, photosensitizer, and co-catalyst, respectively. Therein, the ionic liquid moiety was capable of absorbing and activating high quantities of CO₂ *via* covalent interaction.⁸⁵ Some functional ILs were designed to respond to visible light and then combined with the traditional wide bandgap semiconductors to expand the light response range. For instance, a four-site citrazinate-based ionic liquid, [P₄₄₄₄]₃[p-2,6-O-4-COO], was designed and exhibited super-high CO₂ adsorption capacity. Interestingly, the generated IL-CO₂ complex showed visible light absorption. Thus, [P₄₄₄₄]₃[p-2,6-O-4-COO] could serve as a photosensitizer after adsorbing CO₂ when it was

immobilized on the surface of TiO₂ nanoparticles. Under visible light irradiation, the photogenerated electrons transferred from the CB of the IL-CO₂ complex to the CB of TiO₂, where the activated CO₂ molecules were reduced with trapping protons, thereby producing CH₄.⁸⁷ Recently, Co(Salen)-based photosensitive poly(ionic liquid)s that were composed of heptazine, pyridine ions, and Co(Salen) were designed and synthesized as photocatalysts for syngas production from CO₂. The synthetic process for Co(Salen)-PIL is illustrated in Fig. 6C. In this PIL, heptazine acted as a photosensitization component, while Co(Salen) units served as potent CO₂ activation sites as well as pyridine ions acted as bridges to facilitate electron transfer from heptazine to the Co(Salen) active site. Consequently, the Co(Salen)-PIL displayed excellent photocatalytic activity with a syngas production rate of 168.1 mmol g⁻¹ h⁻¹.⁸⁶

In CO₂ photoreduction, realizing the *in situ* simultaneously capture, enrichment, and conversion of CO₂ is the ultimate goal of developing a highly efficient photocatalytic system. So far, some positive attempts have made progress.⁸⁸ The loading of [Bmim][NTf₂] introduced more active sites into TiO₂ nanotubes, and strengthened the adsorption of CO₂, thereby promoting the generation of acetic acid in the CO₂ photoreduction reaction.⁸⁹ Amine-based functionalized ILs possess stronger CO₂ capture performance compared with other ILs. Thus, [APmim][BF₄] was chosen to modify Bi₂MoO₆ (BMO) for the capture and conversion of CO₂. The introduction of ILs not only promoted the formation of highly dispersed Cu nanodots, but also decreased the activation energy of generating CO₂ intermediates and the overpotential of the photosynthesis reaction of liquid products. As a result, the yield of the liquid products on IL-modified Cu/

BMO increased by 17.80 times compared with BMO.⁹⁰ Similarly, CO₂ molecules were quickly adsorbed by the induction of amino groups in 3-aminopiperidine hydrochloride ([3APip]Cl) modified titanium silicalite-1 (TS-1) molecular sieve nanosheets. Furthermore, [3APip]Cl could capture photoexcited electrons to boost the conversion of carboxyl intermediate to CO and CH₄, thereby enabling the hybrid catalyst to achieve 83% selectivity of CO production.⁹¹ Interestingly, a three-in-one photocatalytic system of CO₂ enrichment, CO₂ reduction and H₂O oxidation sites, which are composed of [Emim][BF₄] and the Zn-Salen-based covalent organic framework (Zn-S-COF), was constructed for dilute CO₂ reduction. As a result, the delicately designed photocatalyst showed a visible-light-driven CO₂-to-CO conversion rate of 105.88 $\mu\text{mol g}^{-1} \text{h}^{-1}$ under a diluted CO₂ (15%) atmosphere. Natural sunlight also drove the diluted CO₂ reduction rate to 126.51 $\mu\text{mol g}^{-1} \text{h}^{-1}$ in 5 h. Therein, the loaded ILs facilitated the mass-transfer of CO₂, providing an enriched CO₂ atmosphere.⁹² In addition, some ILs exhibited other intriguing functions. For example, the 1-octyl-3-methylimidazolium bromide ([Omim]Br) IL could serve as a surface passivation ligand and precursor to mediate the growth and stabilize CsPbBr₃ metal halide perovskites for CO₂ photoreduction. Moreover, with the assistance of [Omim]Br as the activator of CO₂, the CsPbBr₃ provided a high evolution rate of 24.56 and 6.66 $\mu\text{mol g}^{-1} \text{h}^{-1}$ for CO and CH₄, respectively.⁹³

In conclusion, IL-functionalized photocatalysts demonstrate several distinct advantages in comparison to IL-free photocatalysts: (i) ILs could enable stronger CO₂ capture and facilitate CO₂ activation in CO₂ conversion. (ii) ILs could combine with a semiconductor as an active component to accept the photo-generated electrons, boosting the charge separation. (iii) The electronic structure of the semiconductor could also be tuned by the modification of ILs, altering the reaction path or lowering the reaction energy barrier. (iv) The robust chemical bonding between ILs and semiconductors could also affect the stability of the heterojunction photocatalysts. Notably, various types of ILs, especially those modified with different functional groups, demonstrate unique CO₂ adsorption and catalytic properties in CO₂ photoreduction. Optimizing the molecular structure of ILs is a promising way to further enhance their catalytic efficacy.

Conclusion and outlook

This minireview provides a comprehensive overview of recent advancements in ionic liquid (IL)-assisted photocatalytic CO₂ reduction. ILs demonstrate multifaceted roles in enhancing CO₂ photoreduction efficiency, including acting as templates for photocatalyst synthesis, solvents to enhance CO₂ capture, and cocatalysts to promote charge separation.

Despite the significant advantages ILs offer in CO₂ adsorption and conversion, IL-assisted artificial photosynthesis for generating renewable solar fuels from CO₂ and water remains in its nascent stages. The primary challenges include:

(1) Inadequate light harvesting capability: most ILs exhibit relatively weak light absorption properties, limiting their effective utilization in photocatalytic reactions, particularly for

long-wavelength light. There is a critical need for the molecular design of ILs to enhance their light absorption characteristics.

(2) Construction of IL-semiconductor hybrid photocatalysts: synthesizing IL-semiconductor hybrid photocatalysts with superior charge separation properties remains a formidable technical challenge. Researchers must develop effective methods to achieve optimal interactions at the material interfaces, and the electron transfer path is still unclear.

(3) Low solar conversion efficiency: current photocatalytic systems often have limited solar-to-fuel conversion efficiency and suboptimal product selectivity. This restricts their practical application potential. Researchers must optimize specific reaction pathways to enhance product selectivity and deposition reaction efficiency.

(4) Design of separated redox sites: although ILs can improve reaction conditions, effectively creating spatially separated redox sites remains an unresolved issue. This is crucial for enhancing reaction selectivity and efficiency.

Given these challenges, designing robust chemical bonds between ILs and semiconductors, as well as developing spatially separated reaction sites, can effectively address existing obstacles. Future research directions include:

(1) Optimizing IL molecular structures: exploring different types of ILs to identify those with greater potential for efficient photocatalysis. In short, designing photoactive organic cations and catalytically active anions is a promising way to synthesize efficient IL-type photocatalysts.

(2) Integrating multifunctional materials: combining ILs with various semiconductor materials to form photocatalytic systems with synergistic effects, thereby enhancing solar conversion efficiency and catalyst selectivity. Particularly for the production of C₂⁺ hydrocarbons, it is essential to engineer active sites on both the ILs and semiconductors individually. The spatially separated redox sites, combined with their interfacial interaction, offer an ample supply of active sites and an efficient charge transfer pathway, which are critical for facilitating the sluggish C-C coupling reaction.

(3) Developing an integrated reaction system: in the future, it is a promising way to employ the photocatalytic technique to purify flue gases. As displayed in Fig. 7, the photocatalytic membrane reactor, which is assembled from IL-modified

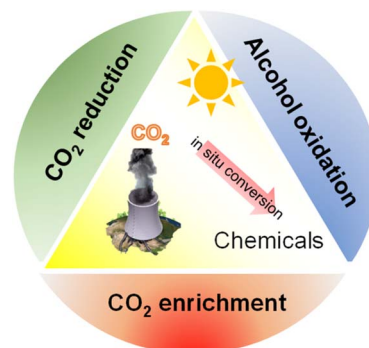


Fig. 7 Proposed strategy for the future photocatalytic CO₂ reduction application.

semiconductors, is a suitable candidate to treat the flue gases. This is attributed to the capability of ILs to capture and enrich dilute CO₂, significantly enhancing CO₂ mass transfer and facilitating the kinetics of the oxidation half-reaction, thereby elevating the overall reaction efficiency.

In conclusion, ILs exhibit unique advantages in photocatalytic CO₂ reduction. However, achieving a highly efficient and selective photocatalytic CO₂ reduction reaction remains a subject of further research and development. We believe that this timely review gives us inspiration to develop more efficient photocatalysts for the generation of solar fuels.

Data availability

No primary research results, software or code have been included and no new data were generated or analysed as part of this review.

Author contributions

Dr Yun Zhao: conceptualization, writing-original draft. Prof. Huihu Wang: funding acquisition, supervision, writing-review & editing. Dr Lei Zeng: conceptualization, resources, writing-original draft. Prof. Limin Huang: funding acquisition, project administration, writing-review & editing.

Conflicts of interest

There are no conflicts to declare.

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